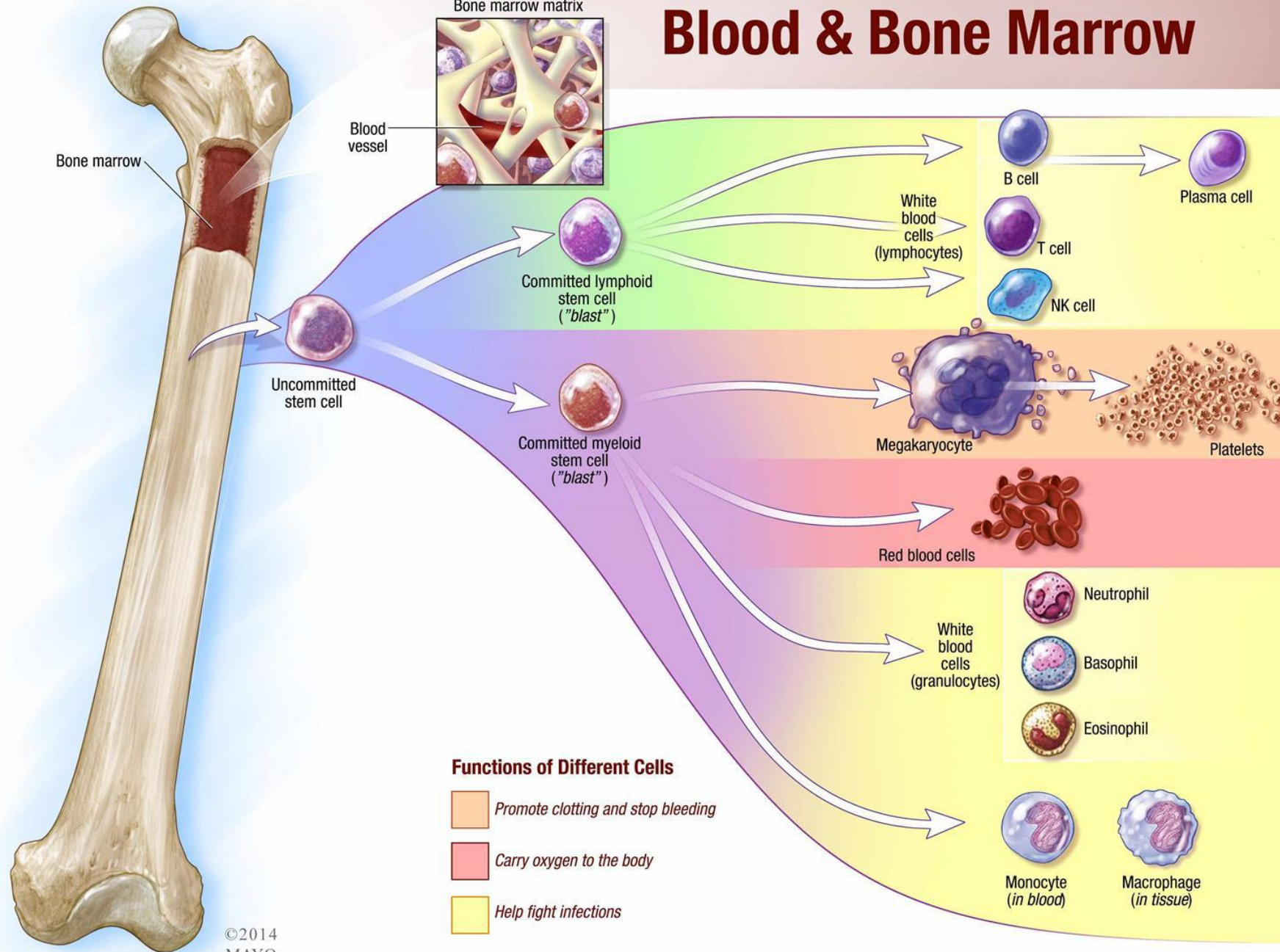




Hematopoiesis

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Blood & Bone Marrow



Hematopoiesis

- Hemato: blood, poiesis: formation.
- Is a continuous, regulated process of blood cells production that includes cell renewal, proliferation, differentiation, and maturation.
- These processes result in the formation, development, and specialization of all of the functional blood cells that are released from the bone marrow to the circulation.
- Hematopoiesis aims to maintain an adequate number or count of blood cells to maintain their functions (gas exchange, defense and immunity, and hemostasis).

➤ **Haemopoiesis:**

-Means blood cell formation

➤ **Site of hemopoiesis:**

Fetus (0-2 months Yolk Sac)
(2-7 months Liver, spleen)
(5-9 months Bone marrow)

Infants and child (bone marrow)

Adults (vertebrae, ribs, sternum, skull, sacrum and pelvis, the proximal end of the femur (central Skelton)) because of progressive fatty replacement.

➤ **Hematopoiesis includes:**

- 1) RBCs (Erythrocytes) production: Erythropoiesis.
- 2) WBCs (Leukocytes) production: Leukopoiesis.
- 3) Platelets (Thrombocytes) production: Thrombopoiesis.

Hematopoiesis Phases

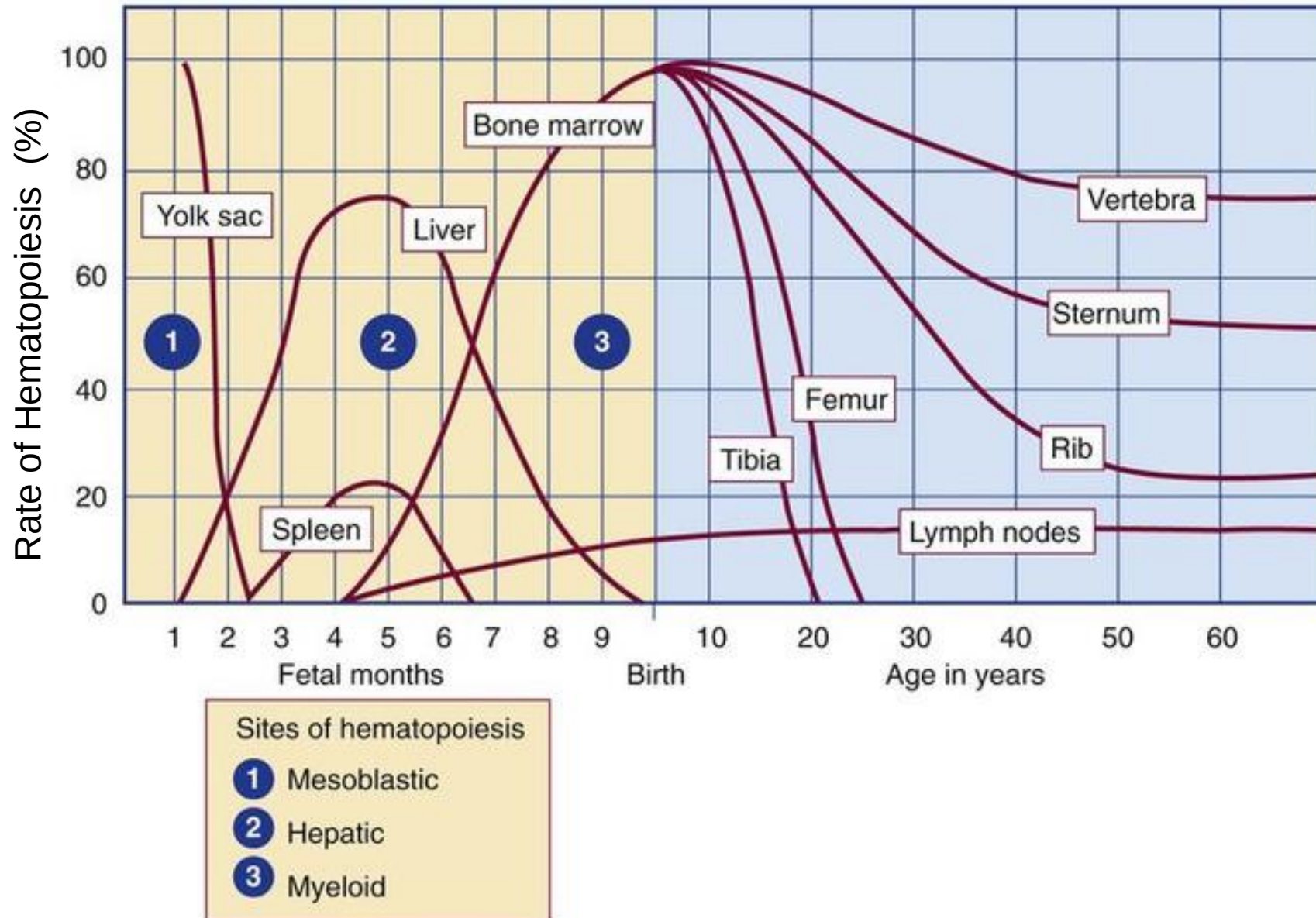
- 1) Mesoblastic phase (Primitive hematopoiesis):** Hematopoiesis begins in the yolk sac during the first month of embryogenesis (until the 6th week).
- 2) Hepatic phase:** From (5-7) weeks until (6-7) months of pregnancy, hematopoiesis occurs in the liver and spleen, but the liver is the main organ of hematopoiesis and they continue to produce blood cells until 1-2 weeks after birth.

3) Medullary or Myeloid phase: From 6-7 months of fetal life, bone marrow (BM) becomes the primary site of hematopoiesis and continues to be the only site of hematopoiesis in childhood and adult life in normal conditions (**intramedullary hematopoiesis**).

➤ **2+3: definitive hematopoiesis.**

➤ Hematopoiesis may resume in the liver and spleen after birth in some conditions when needed (**extramedullary hematopoiesis**).

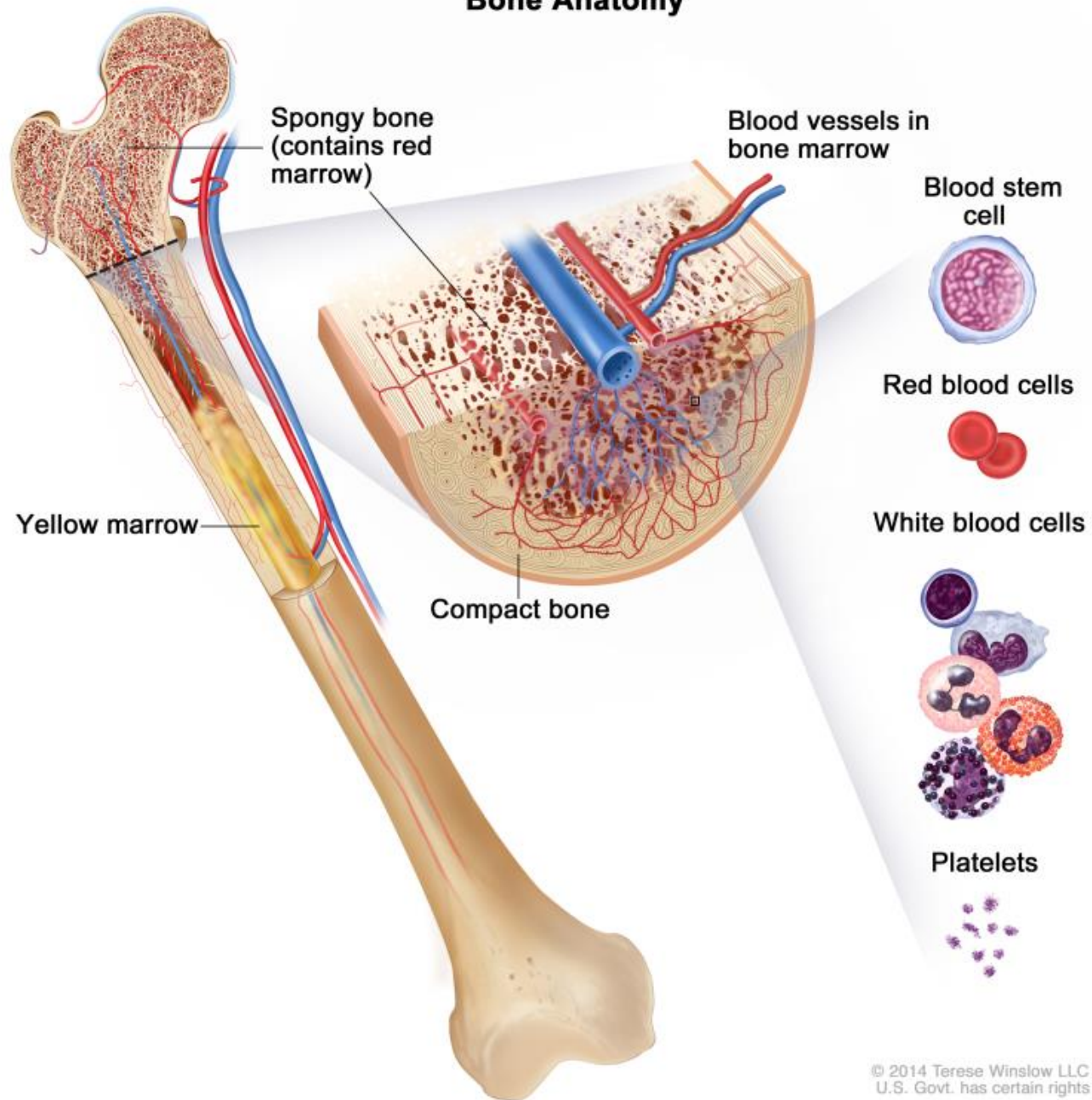
Sites of Hematopoiesis by Age



Bone Marrow

- **Bone marrow** is the flexible tissue in the interior of bones.
- On average, bone marrow, one of the largest organs in the body, constitutes 4% of the total body mass of humans.
- Normal bone marrow contains two major components:
 - 1) “**RED MARROW**”, hematopoietically active marrow consisting of the developing blood cells and their progenitors,
 - 2) and “**YELLOW MARROW**”, hematopoietically inactive marrow composed primarily of adipocytes (fat cells).
- Blood cells are produced mainly in the “**RED MARROW**”.

Bone Anatomy



- During infancy and early childhood, all the bones in the body contain primarily red (active) marrow.
- Between 5 and 7 years of age, adipocytes become more abundant and begin to occupy the spaces in the long bones previously dominated by active marrow.
- The process of replacing the active marrow with adipocytes (yellow marrow) during development is called **retrogression** and eventually results in restriction of the active marrow in the adult to the sternum, vertebrae, scapulae, pelvis, ribs, skull, and proximal portion of the long bones (femurs and humeri).

- Hematopoietically inactive yellow marrow is scattered throughout the red marrow so that in adults, there are approximately equal amounts of red and yellow marrow in these areas (50:50).
- Yellow marrow is capable of returning to active marrow in cases of increased demand on the bone marrow, such as in excessive blood loss or hemolysis.
- For the average adult, the bone marrow produces $\sim 5 \times 10^{11}$ cells per day.
- BM is the site of differentiation and maturation of myeloid-derived cells and the primary site of origin of lymphocytes.

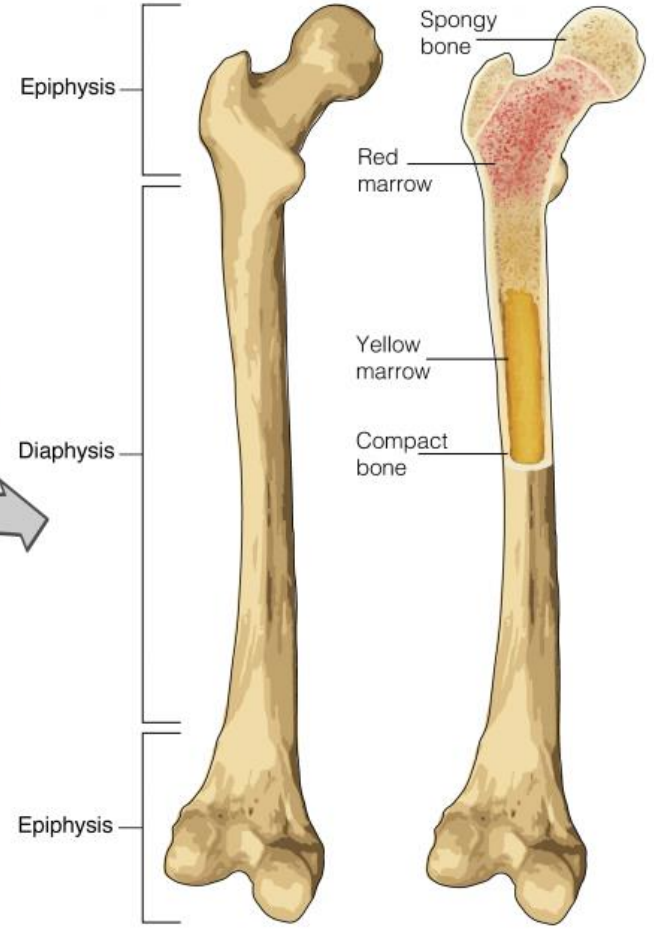
- The bone marrow contains hematopoietic cells, stromal cells, and blood vessels (arteries and veins).
- The stroma of the bone marrow is all tissue not directly involved in the marrow's primary function of hematopoiesis.
- Stromal cells include endothelial cells, adipocytes (fat cells), macrophages and lymphocytes, osteoblasts, osteoclasts, and reticular adventitial cells (fibroblasts).
- Yellow bone marrow makes up the majority of BM stroma.
- Stroma is indirectly involved in hematopoiesis since it provides the suitable hematopoietic microenvironment that facilitates hematopoiesis.

Red Marrow
Yellow Marrow



Infant <1 yr

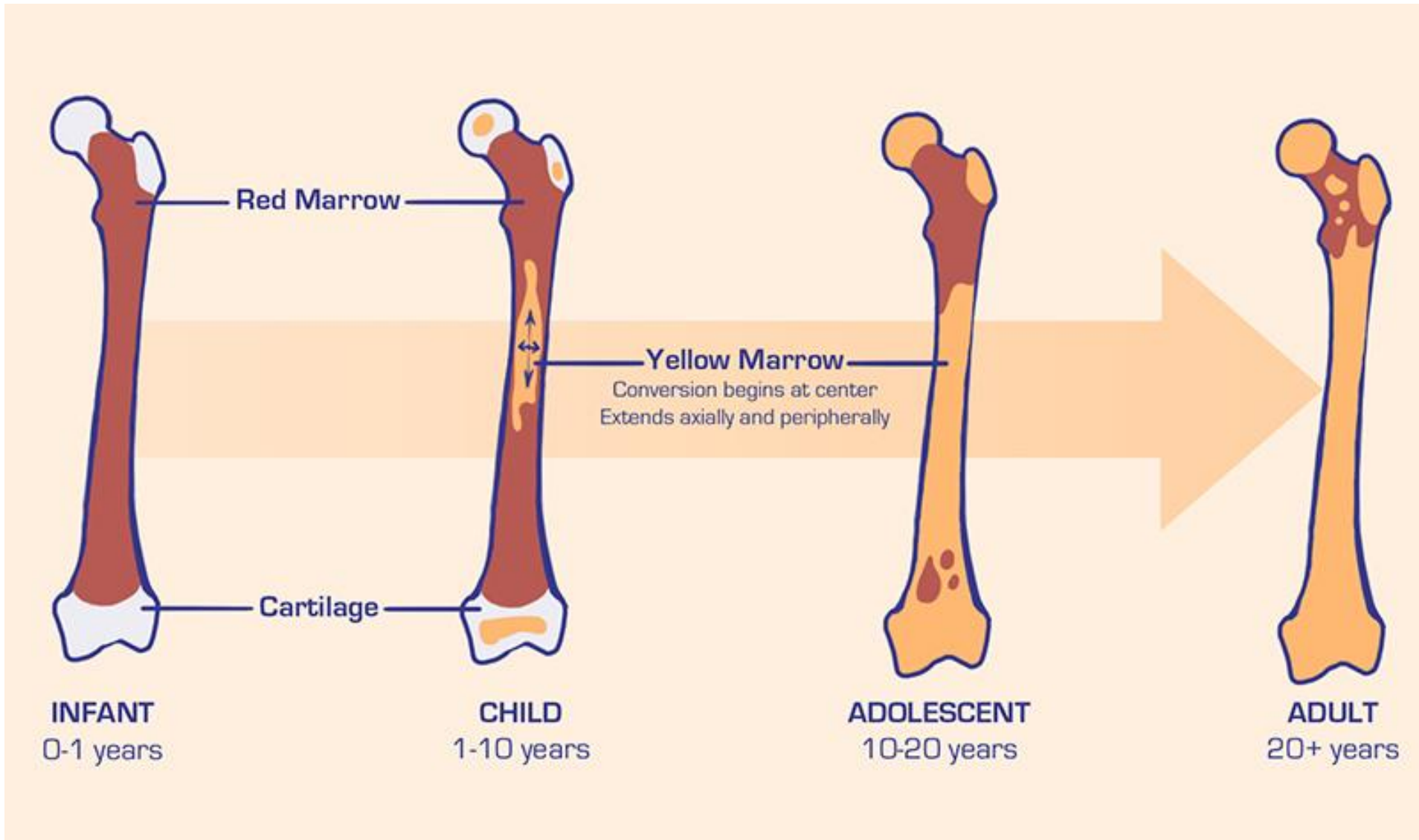
Adult 25+ yrs



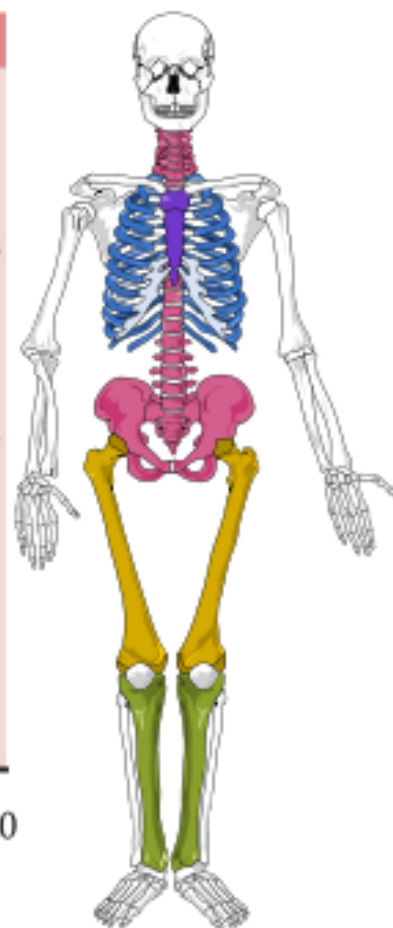
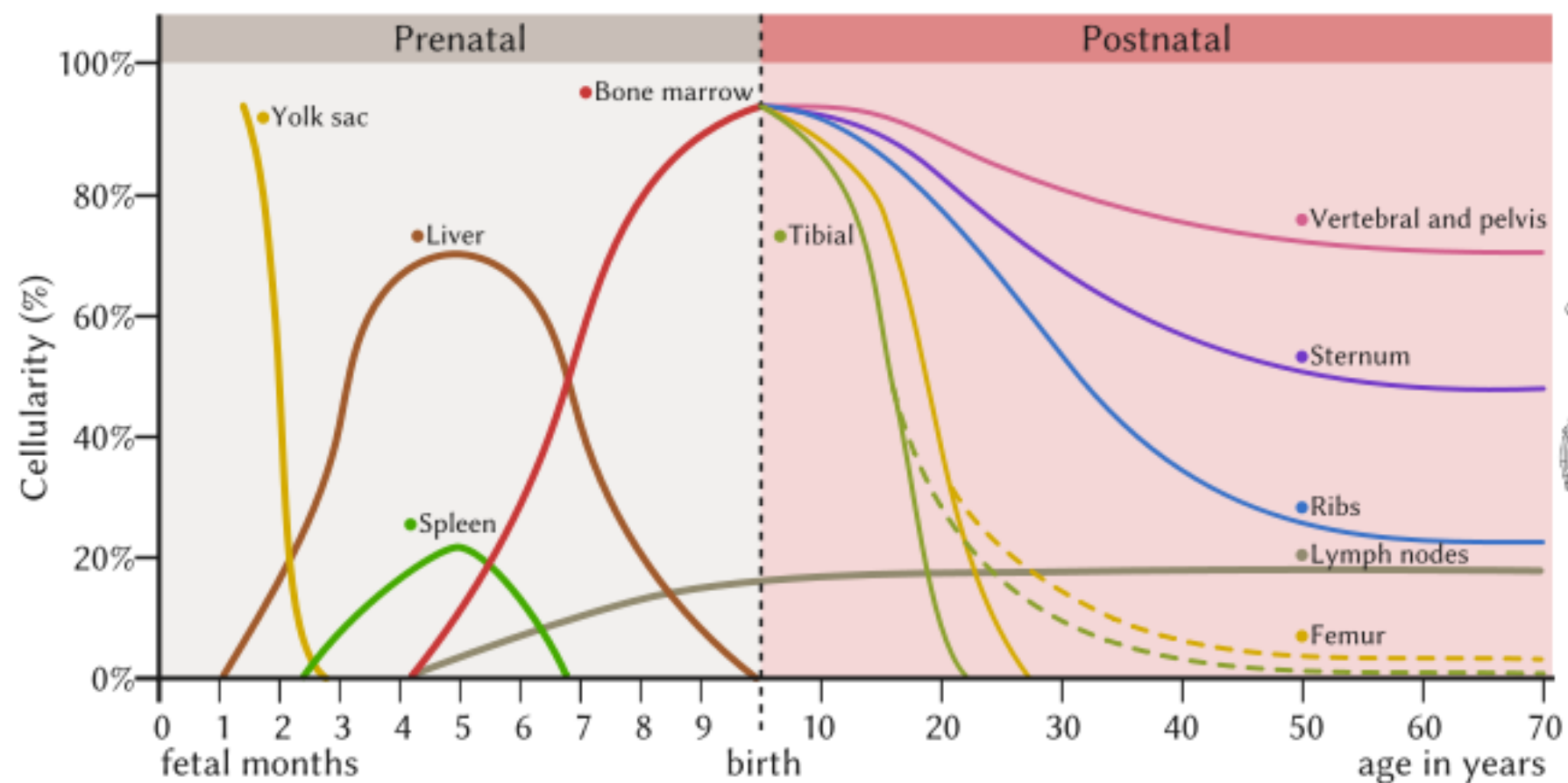
Left femur
posterior view

Left femur
coronal section

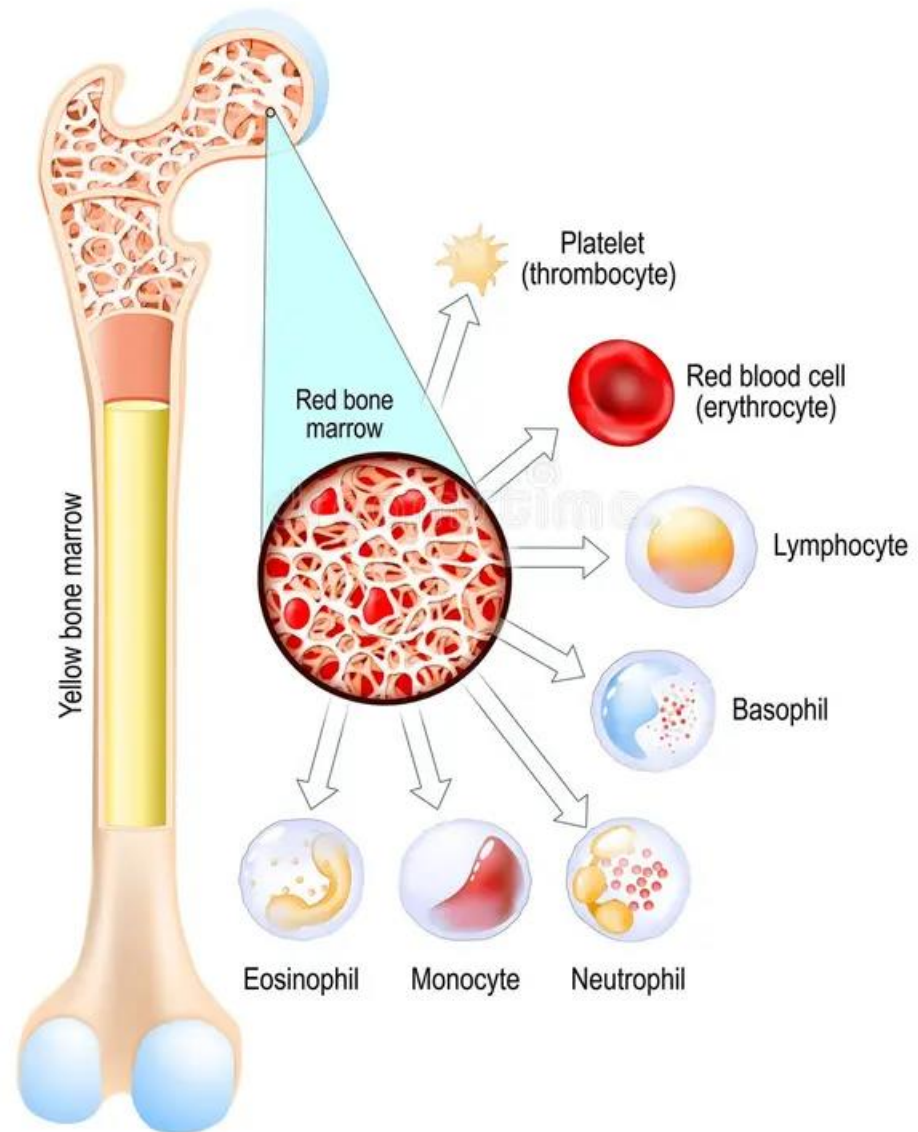
Normal bone marrow conversion



HEMATOPOIESIS•



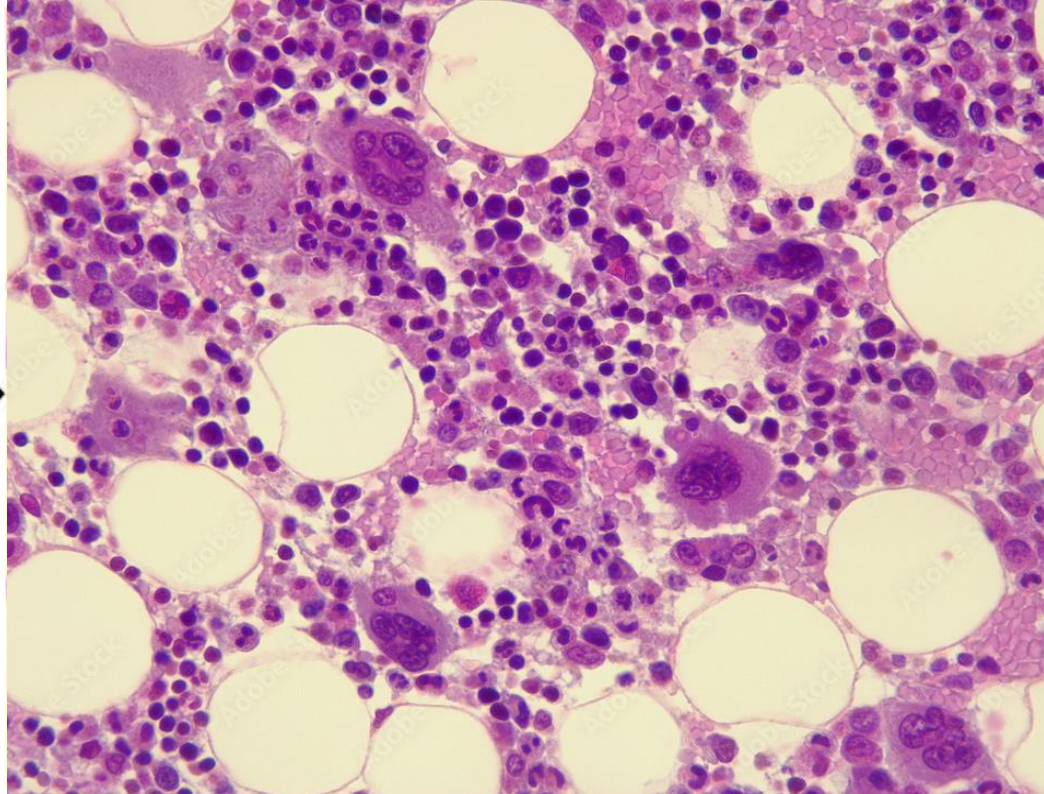
Bone marrow



Bone marrow

Blood

Senescent cells
Apoptotic cells
Lipoproteins



Blood

Blood cells

Granulocytes	120×10^9 /day
Erythrocytes	200×10^9 /day
Platelets	150×10^9 /day
Lymphocytes	20×10^9 /day

Hematopoiesis
(Cell differentiation and proliferation)

	RBC	WBC	Platelet
Appearance	Biconcave, circular	Irregular shape	Irregular shape
Site of production	Bone Marrow	Lymphocytes – Lymph nodes Phagocytes – Bone marrow	Bone marrow
Life span	120 days	10 days	6 days
Presence of Hb and nucleus	No nucleus Hb present	Nucleated Hb absent	No nucleus Hb absent
Generation rate (Cells/day)	2×10^{11}	2×10^{10}	1.5×10^{11}

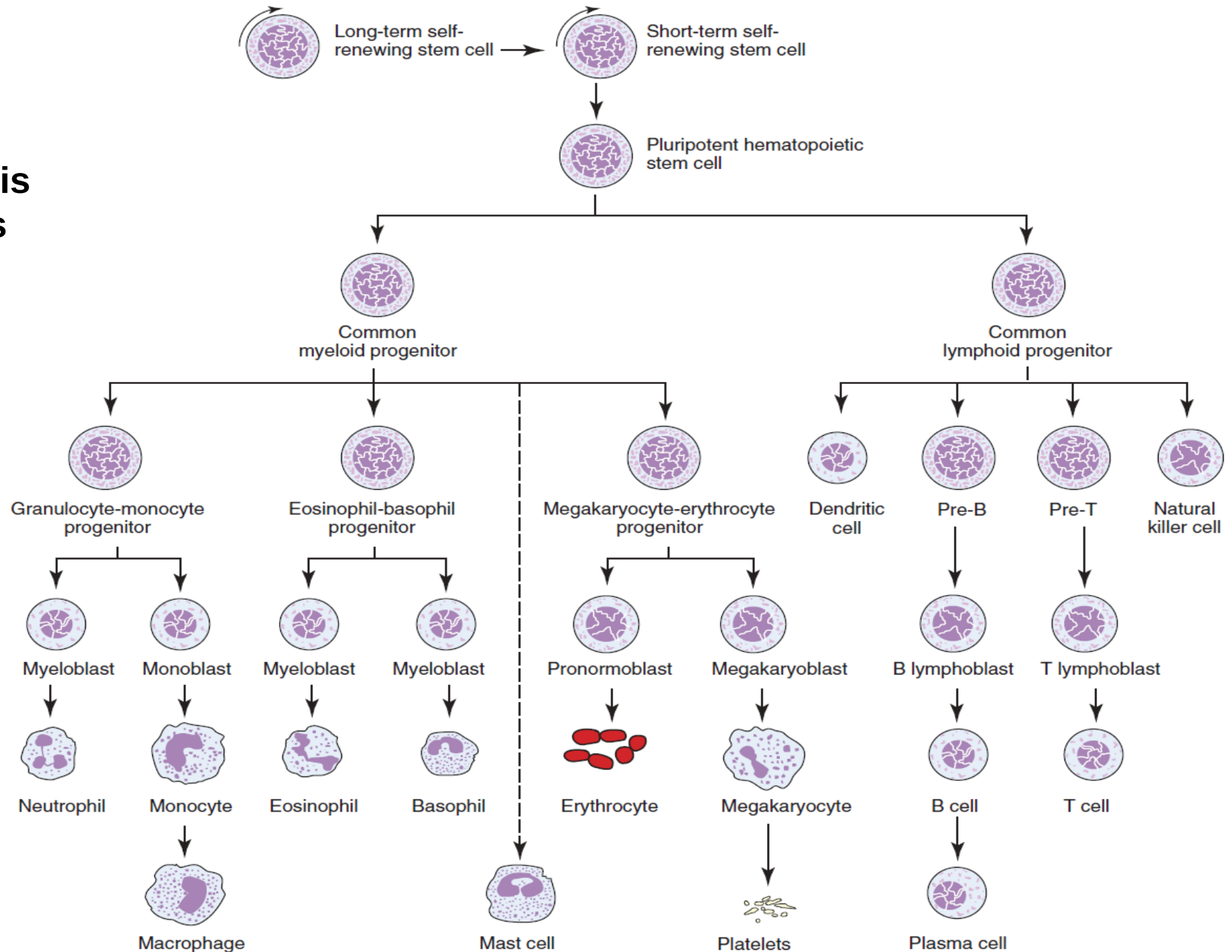
Stem cells

- Stem cells are undifferentiated unspecialized biological cells that can differentiate into specialized cells and can divide (through mitosis) to produce more stem cells.
- Stem cell has the characteristics of :
 - 1) Self-renewal.
 - 2) Potency: the capacity to differentiate into specialized cell types
 - 3) Mitotically quiescent (the cells retain the ability to re-enter cell proliferation).
 - 4) Able to repopulate a bone marrow after treatment.
- Can be totipotent (All types of cells) or pluri/ multipotent (Many types of cells).
- Stem → progenitor → precursor → adult → mature

- Despite the limited numbers of HSCs in the bone marrow, 6 billion blood cells per kilogram of body weight are produced each day for the entire life span of an individual.
- Most of the cells in normal bone marrow are precursor cells at various stages of maturation.
- HSCs are directed to one of three possible fates: self-renewal, differentiation, or apoptosis.
- When the HSC divides, it gives rise to two identical daughter cells.
- Both daughter cells may follow the path of differentiation, leaving the stem cell pool (**symmetric division**), or one daughter cell may return to the stem cell pool, and the other daughter cell may follow the path of differentiation (**asymmetric division**) or undergo apoptosis.

- The undifferentiated HSCs can differentiate into progenitor cells committed to either lymphoid or myeloid lineages.
- These lineage-specific progenitor cells are the **common lymphoid progenitor**, which proliferates and differentiates into T, B, and natural killer lymphocyte and dendritic lineages; and the **common myeloid progenitor**, which proliferates and differentiates into individual granulocytic, erythrocytic, monocytic, and megakaryocytic lineages.
- The resulting limited lineage-specific progenitors give rise to morphologically recognizable, lineage-specific precursor cells.

Diagram of hematopoiesis shows derivation of cells from the pluripotent hematopoietic stem cell.



- The bone marrow is estimated to be capable of producing approximately 2.5 billion erythrocytes, 2.5 billion platelets, and 1 billion granulocytes per kilogram of body weight daily.
- The determining factor controlling the rate of production is physiologic need.
- HSCs exist in the marrow in the ratio of 1 per 1000 nucleated blood cells.
- They are capable of many mitotic divisions when stimulated by appropriate cytokines.

Regulation Of Hematopoiesis

- Production of blood cells is highly regulated and balanced.
- The hematopoietic system consists of cells, tissues, and organs that continuously cooperate to maintain the proper amount of blood cells.
- The hematopoietic system consists of the bone marrow, liver, spleen, lymph nodes, and thymus.
- All these tissues and organs are involved in the proliferation, production, differentiation, maturation, and destruction of blood cells.
- Under the influence of certain growth factors such as colony-stimulating factors (CSF) and interleukins (IL), stem cells divide, differentiate, and mature to form the mature cellular elements of the blood.

Cytokines and Growth Factors

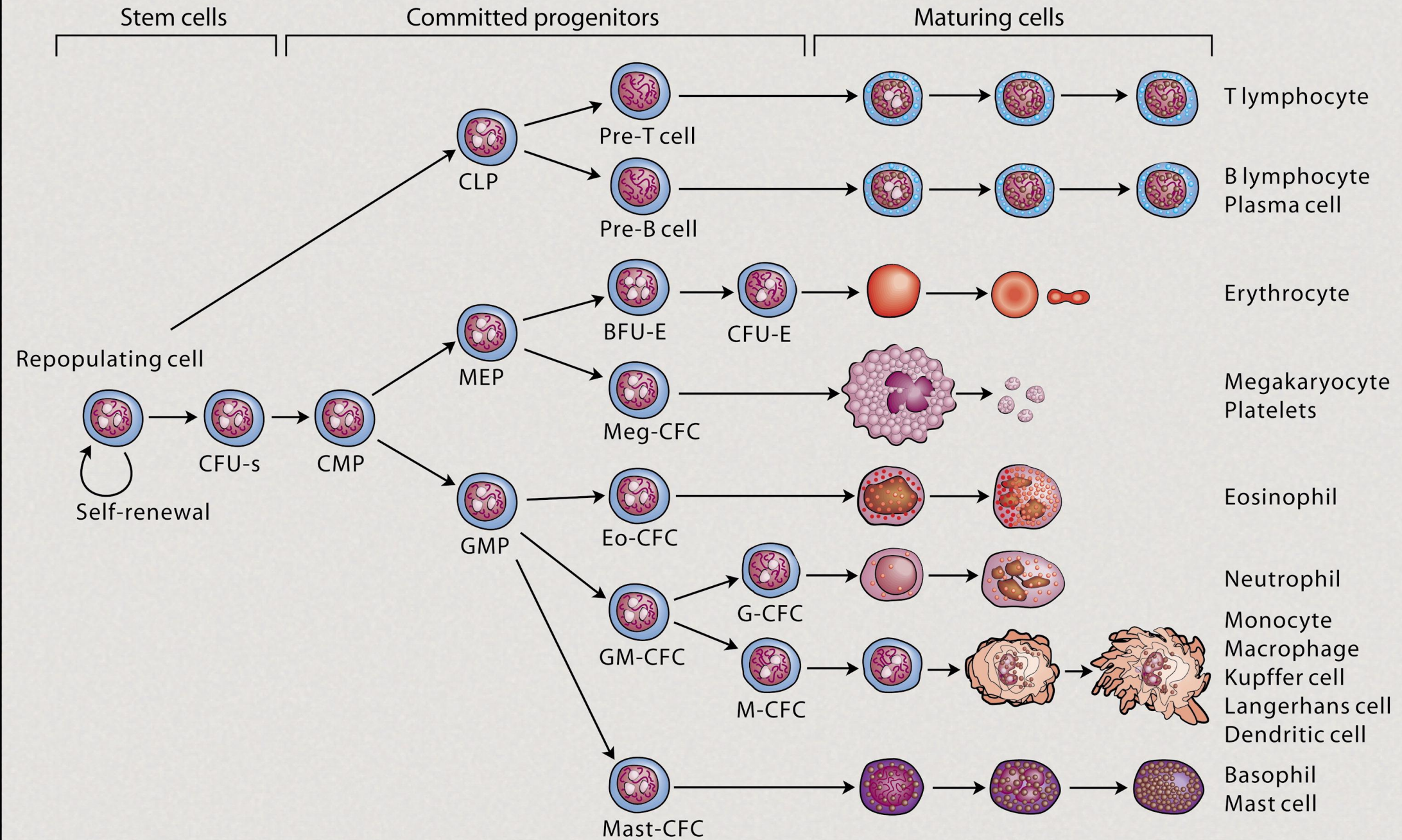
- A group of specific glycoproteins called hematopoietic growth factors or cytokines regulate the proliferation, differentiation, and maturation of hematopoietic precursor cells.
- Cytokines are a diverse group of soluble proteins that have direct and indirect effects on hematopoietic cells.
- The terms cytokine and growth factor are often used synonymously; cytokines include interleukins (ILs), lymphokines, monokines, interferons, chemokines, and colony-stimulating factors (CSFs).
- Cytokines are responsible for stimulation or inhibition of the production, differentiation, and trafficking of mature blood cells and their precursors.

- All blood cells are derived from **pluripotent (multipotential) hematopoietic stem cells (HSC)** found in the BM.
- The progeny of these cells are capable of giving rise to all the different lines of mature blood cells: erythrocytes, granulocytes, monocytes, lymphocytes, and megakaryocytes (platelets).
- The pluripotent stem cells are capable of self-renewal.

- They are rare in the bone marrow (~1 per 1,000 nucleated marrow cells) and cannot be recognized morphologically.
- The majority of pluripotent stem cells at any given time are in the resting (Go) phase of the cell cycle.
- Pluripotent hematopoietic stem cells give rise to **committed progenitor cells**.

- This is a multi-step process in which the cells become sequentially more committed to a specific lineage.
- These committed progenitor cells are given various names, such as **colony-forming unit (CFU)** or **burst-forming unit (BFU)**.
- The initial differentiation step is into either **common myeloid progenitor (CMP)** or **CFU-GEMM** (GEMM = granulocytes, erythrocytes, monocytes/macrophages, and megakaryocytes) or **common lymphoid progenitor (CLP)** or **CFU-L** (L = lymphoid).

- The CFU-GEMM gives rise to the **CFU-GM** (granulocyte-macrophage), **BFU-E** (erythroid), and **CFU-Mk** (megakaryocyte).
- Each committed progenitor cell gives rise to a thousand or more mature blood cells.
- All differentiation and maturation steps of blood cells occur in the BM, except for the T-lymphocytes which occur in both BM and thymus.



Hematopoiesis Hierarchy

- Describe the process of transition from SCs to the progenitor cells.
- Progenitor cells: Those are uni-potential or multi-potential cells with little self-renewal capacity.
- Precursor cells exhibit certain recognized morphologic, immunological, and cytochemical features with **no** self-renewal capacity.
- Differentiation: specialization with unique characteristic features.
- Maturation: ends with the generation of fully functional elements.

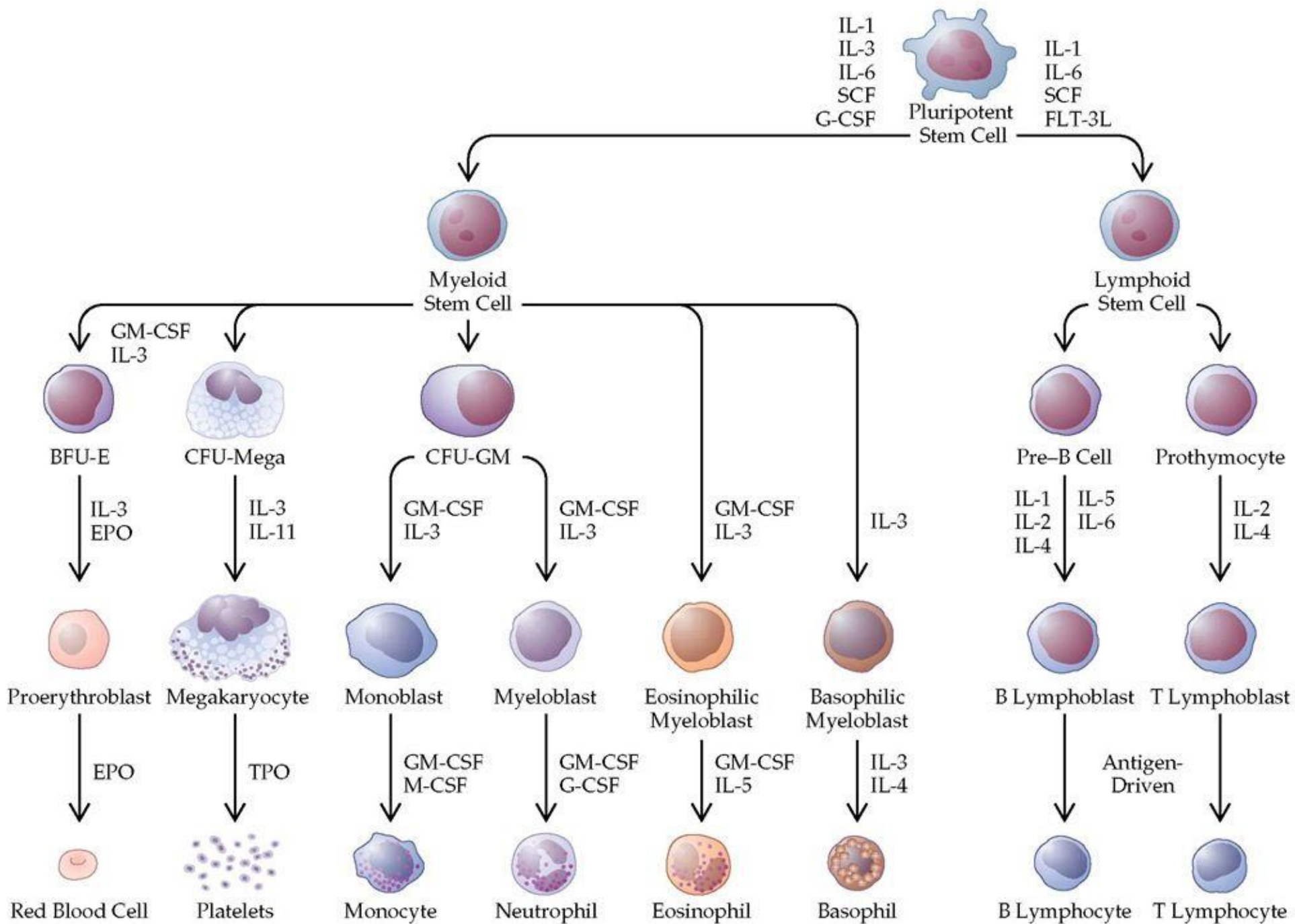
Hematopoietic growth factors

- Hematopoietic growth factors are proteins or glycoproteins that regulate the:
 - 1) Proliferation,
 - 2) differentiation,
 - 3) maturation and
 - 4) prevention of apoptosis of hematopoietic precursors.

- They are termed as interleukins (IL) or colony-stimulating factor (CSF).

Hematopoietic growth factors

- 1) Erythropoietin (Epo)
- 2) Granulocyte-colony stimulating factor (G-CSF)
- 3) Granulocyte / Macrophage-colony stimulating factor (GM-CSF)
- 4) Interleukin 1 (IL-1)
- 5) Interleukin 3 (IL-3)
- 6) Interleukin 6 (IL-6)
- 7) Macrophage-colony stimulating factor (M-CSF)
- 8) Stem-cell factor (SCF)
- 9) Thrombopoietin (Tpo)



Bone marrow examination

1. Aspiration:

- A needle is inserted into the marrow and a liquid sample is obtained.
- Spread on a slide and stained with Romanowsky technique.
- Details of developing cells can be examined. As well as, the proportion of different cell lines, and the presence of cells foreign to the marrow.

2. Trephine:

- Provides solid core of bone.
- Examined as a histological specimen.
- Less valuable than aspiration but provides a panoramic view of the marrow. Important to see overall marrow architecture, cellularity, and the presence of fibrosis.

Summary

- Hematopoiesis is the process of blood cell formation.
- Hematopoiesis begins in the yolk sac, through liver and spleen and finally in the BM.
- The Hematopoietic system includes the BM, liver, spleen, thymus, and lymph nodes.
- The BM is the primary site of hematopoiesis at birth and throughout life.
- Intramedullary hematopoiesis occurs in the BM, while extramedullary hematopoiesis occurs in the liver and spleen.
- BM is red (active) in infants and children in all bones, while the red marrow is decreased and replaced by yellow (fat) marrow and restricted to the axial skeleton bones and the proximal end of long bones in adults.

- Liver and spleen and yellow marrow can resume their hematopoietic activity in some conditions.
- Stem cell is the unspecialized undifferentiated cell that has the capacity to give specialized differentiated cell.
- Hematopoiesis is a very complex regulated process that is regulated through a number of growth factors or cytokines called interleukins (IL) and colony stimulating factors (CSF) and other factors.
- All differentiation and maturation step of blood cells occur in the BM, except for the T-lymphocytes which occur in both BM and thymus.